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# Evaluating Physics-Informed Learning for Fluid Dynamics


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## Which is the better way to teach a network fluid flow: simulation data or physics?

A trained neural network predicts a flow in milliseconds instead of running a slow simulation. A data-driven network (data NN) learns from simulation results, so a simulation must run first. A physics-informed network (PINN) [3] learns only from the flow equations: no data, but more computing. A hybrid learns from both.

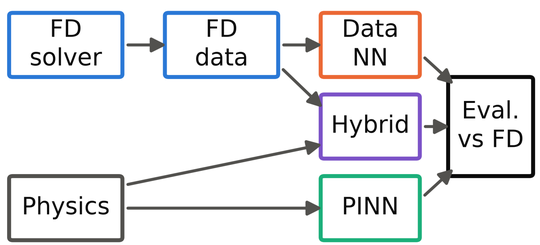
RQ1 Which is more accurate?

RQ2 Which obeys the physics better?

RQ3 Which is more reliable across seeds and flow speeds?

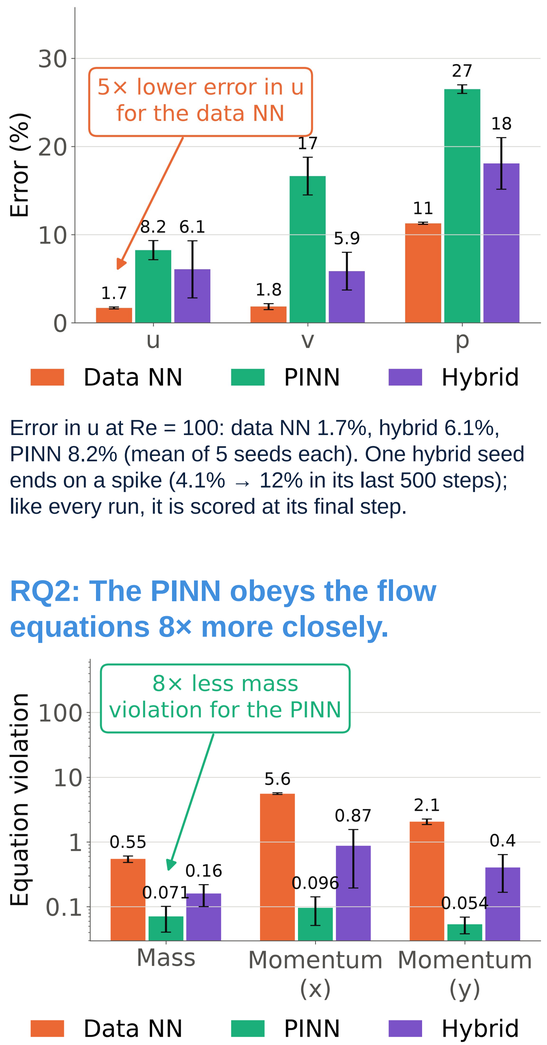
RQ4 What does each cost?

## All three networks were built and scored the same way; only what they learn from differs.




Test flow: fluid in a box driven by a sliding lid. Reference: a finite-difference (FD) simulation that matches published results [1, 2]. All three: 17,091 parameters, 20,000 training steps, 3-5 seeds, scored on points and times never seen in training.

## RQ1: The data NN matches the simulation 5× more closely.



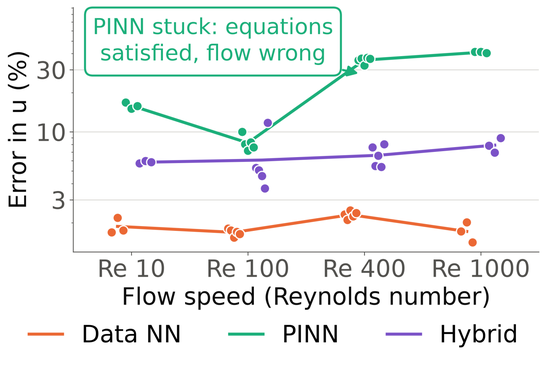

Error in u at Re = 100: data NN 1.7%, hybrid 6.1%, PINN 8.2% (mean of 5 seeds each). One hybrid seed ends on a spike (4.1% → 12% in its last 500 steps); like every run, it is scored at its final step.

## RQ2: The PINN obeys the flow equations 8× more closely.



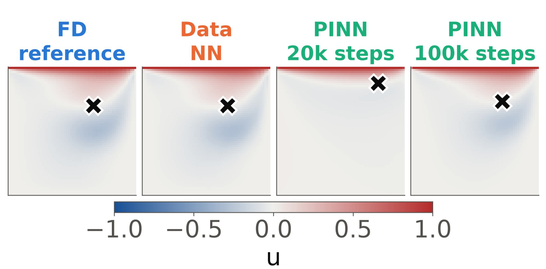

Violation of mass conservation: PINN 0.071, hybrid 0.16, data NN 0.55. Without its physics loss the hybrid's mass violation is 0.45 (error in u 3.0% vs 5.0%, same seeds).

## RQ3: As the flow speeds up, only the PINN breaks down.



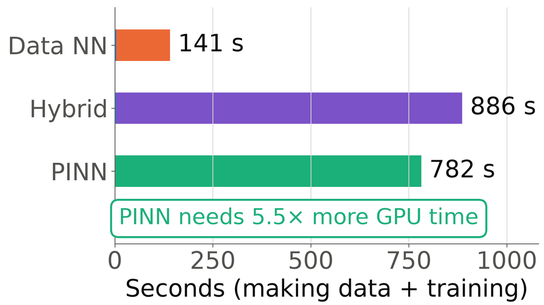

Every seed is a dot. Error in u at Re = 10, 100, 400, 1000: data NN 1.9, 1.7, 2.3, 1.7%; PINN 16, 8.2, 36, 41%; hybrid 5.8, 6.1, 6.6, 7.9%.

## RQ3: At Re = 400 the PINN obeys the equations but needs more training, or a head start, to get the flow right.




Flow at t = 5 (× = vortex). The PINN conserves mass better than the data NN (0.03 vs 0.55) yet puts the vortex by the lid. Error in u: 35% → 14% with 5× more steps (3 seeds); 36% → 15% starting from the trained Re = 100 PINN (5 seeds; a random start is at 21% after the same 40,000 steps). The hybrid gets the flow right (6.6%) and conserves mass far better than the data NN (0.07); trained 5× longer it reaches 4.4% (1 seed).

## RQ4: The PINN needs no data, but 5.5× more training time.




The simulation took only 3.9 s, so data was cheap; the hybrid pays for both. GPU memory: 708 MB (PINN) vs 122 MB (data NN).

## No method wins on everything; each trades accuracy, physics and cost.

- Data NN: most accurate and cheapest, when simulation data already exists.
- PINN: needs no data and obeys the physics best, but costs more and fails as the flow speeds up.
- Hybrid: accurate at every flow speed (7.9% vs 41% for the PINN at Re = 1000) and far closer to the physics than the data NN, but needs data and the most training.
- Reusing a trained network helps: a PINN started from the Re = 100 one more than halved its error at Re = 400.


[1] Marchi, C. H., Suero, R., and Araki, L. K. 2009. The lid-driven square cavity flow: numerical solution with a 1024 x 1024 grid. J. Braz. Soc. Mech. Sci. Eng. 31, 3, 186–198.

[2] Hou, S., Zou, Q., Chen, S., Doolen, G., and Cogley, A. C. 1995. Simulation of cavity flow by the lattice Boltzmann method. J. Comput. Phys. 118, 2, 329–347.

[3] Raissi, M., Perdikaris, P., and Karniadakis, G. E. 2019. Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. J. Comput. Phys. 378, 686–707.



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